



Lecture : 56

LEARNING FOR SUSTAINABILITY



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Learning about Sustainability

Sustainably vs. sustainable development:

- Sustainability as used in 'environmental sustainability', 'economic sustainability' and 'social sustainability' is a one dimensional concept.
- Sustainable development is multidimensional as it integrates, it strives to sustain or even enhance all the dimensions.

- Eggert, R.G(2009) *What sustainability and sustainable development mean in mining, chapter 3,pp 19-32.*



History of sustainable development

- Term first mentioned in a book related to the 1972 UN Stockholm conference on human environment.
- Sustainable development was the solution to the problems of environmental degradation discussed by the Brundtland Commission in the 1987 report *Our Common Future*.

“Sustainable development is development which meets the needs of the present without compromising the ability of future generations to meet their own needs.”

- *Brundtland Commission ,Our Common Future, 1987*



- **Environmental Sustainability:**

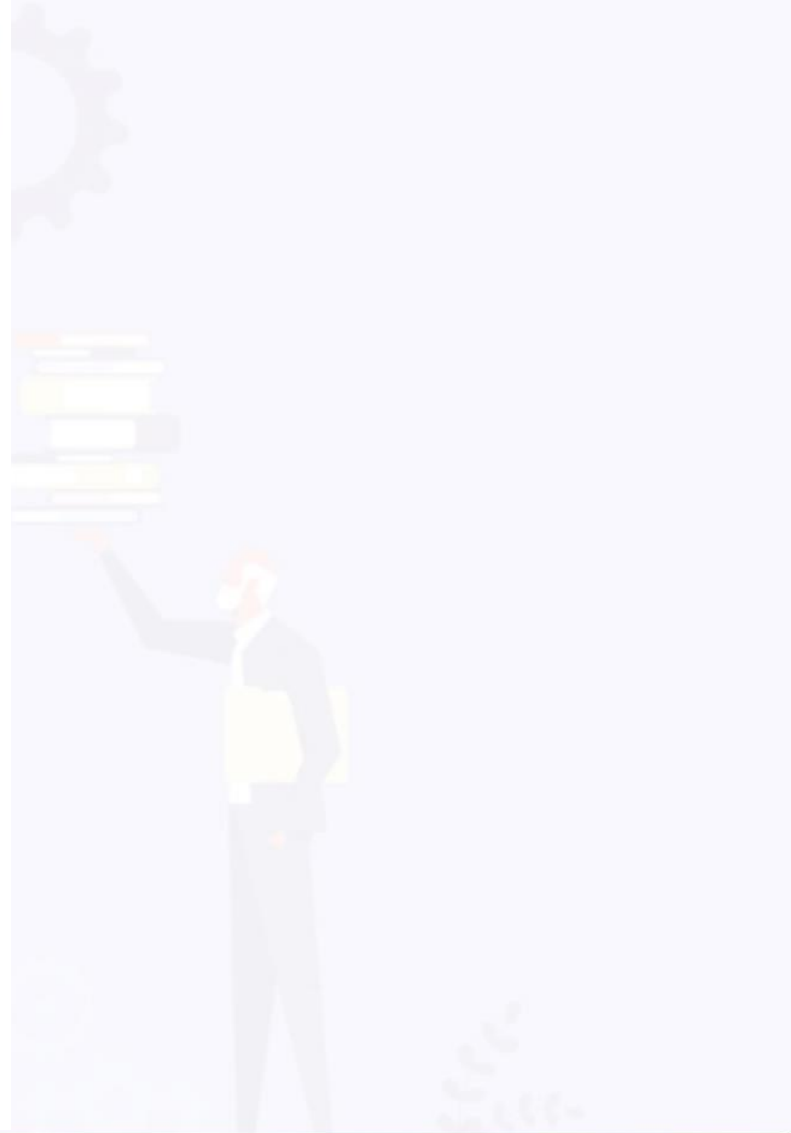
- Environmental sustainability creates and sustains the conditions with which both human and nature can co-exist in a productive harmony with one another while still being able to maintain social and economic requirements.
- ➤ Impacts of processes, products, and services on
 - ✓ air,
 - ✓ water,
 - ✓ land,
 - ✓ biodiversity,
 - ✓ human health,
 - ✓ Ocean environment,
 - ✓ Plantations , etc.



Social Sustainability:

Everything from basic human rights and corporate governance to the more philosophical aspect of an individual's behaviours and attitude toward sustainability.

- Workplace health and safety
- community relations
- employee retention
- labour practices
- business ethics
- human rights
- working conditions etc.
- Social safety and security



Economic Sustainability:

- Productivity,
- Efficiency,
- Revenue generation,
- Alternative usage,
- Resource monitoring,
- Profitability,
- wages and benefits,
- resource use,
- labour productivity,
- job creation,
- expenditures on outsourcing and human capital,
- Capacity development etc.



Primary goals of sustainability:

- The end of poverty and hunger
- Better standards of education and healthcare - particularly as it pertains to water quality and better sanitation
- To achieve gender equality
- Sustainable economic growth while promoting jobs and stronger economies
- All of the above and more while tackling the effects of climate change, pollution and other environmental factors that can harm and do harm people's health, livelihoods and lives.
- Sustainability to include health of the land, air and sea





Social entrepreneurship, defined as solving social problems using market-based methods, has gained in popularity over the past two decades and is increasingly acknowledged as a practice that can create both economic and social value . (Hota et al., 2019)



THANK YOU



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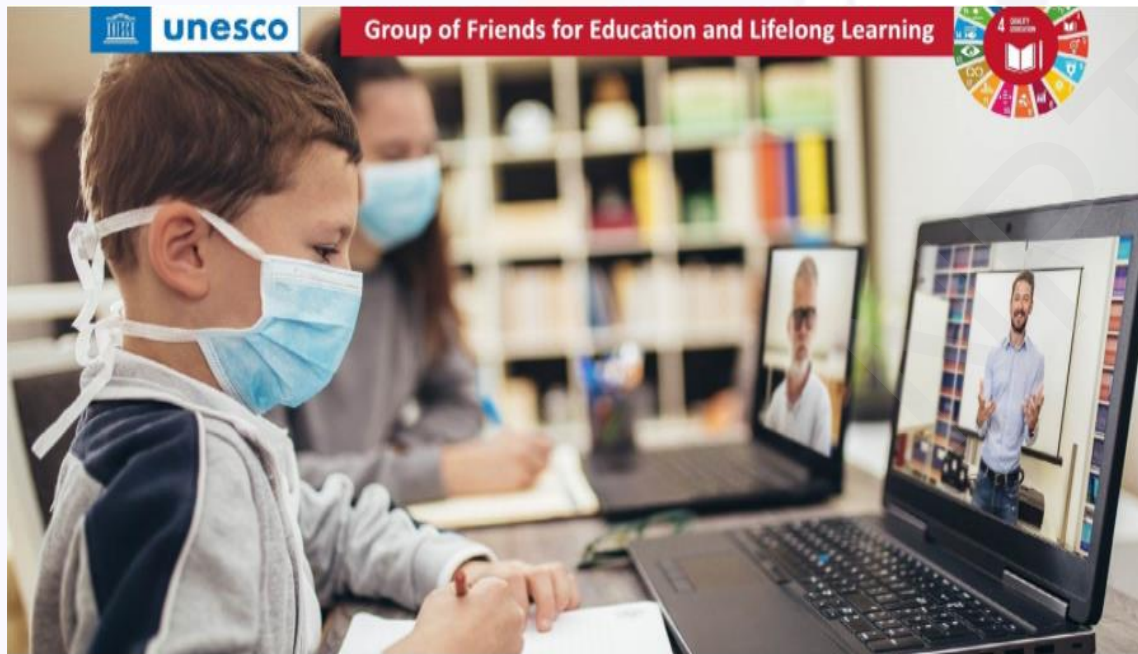


- **United Nations alert that education should be a clear priority**



HESI highlights the role of higher education in building a better world for current and future generations

On 20 October 2020, the Higher Education Sustainability Initiative (HESI) raises the flag of higher education as a driver for sustainable development and inclusive societies on the first day of the Global Education Meeting 2020.



- **Sustainable Development Goal 4** is the education-related goal of the United National 2030 Agenda for Sustainable Development, adopted in September 2015. Its overall aim is: to: “**ensure inclusive and equitable quality education and promote lifelong learning opportunities for all**”.
- **Education 2030 Framework for Action-**
- **Seven Outcome Targets-**
- **4.1 Universal primary and secondary education-** By 2030, ensure that all girls and boys complete free, **equitable and quality** primary and secondary **education** leading to relevant and effective learning outcomes;
- **4.2 Early childhood development and universal pre-primary education-** By 2030, ensure that all girls and boys have **access to quality early childhood** development, care and pre-primary education so that they are ready for primary education



- **4.3 Equal success to technical/vocational and higher education-** By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university
- **4.4 Relevant skills for decent work-** By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship
- **4.5 Gender equality and inclusion-** By 2030, eliminate gender disparities in education and ensure equal access to all levels of education and vocational training for the vulnerable, including persons with disabilities, indigenous peoples and children in vulnerable situations;
- **4.6 Universal youth literacy-** By 2030, ensure that all youth and a substantial proportion of adults, both men and women, achieve literacy and numeracy
- **4.7 Citizenship education for sustainable development -**By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development.



- **Three means of implementation**
- **4.a Effective learning environments-** Build and upgrade education facilities that are child, disability and gender sensitive and provide safe, non-violent, inclusive and effective learning environments for all
- **4.b Scholarships** -By 2030, substantially expand globally the number of scholarships available to developing countries, in particular least developed countries, small island developing States and African countries, for enrolment in higher education, including vocational training and information and communications technology, technical, engineering and scientific programs, in developed countries and other developing countries
- **4.c Teachers and educators-** By 2030, substantially increase the supply of qualified teachers, including through international cooperation for teacher training in developing countries, especially least developed countries and small island developing States



Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

A. One-page summary of *ESD for 2030* framework



EDUCATION FOR SUSTAINABLE DEVELOPMENT Towards achieving the SDGs: *ESD for 2030*

**GLOBAL ACTION
PROGRAMME**
(2015-2019)

ESD for 2030
(2020-2030)

Vision

ESD for 2030 aims to build a more just and sustainable world through strengthening ESD and contributing to the achievement of all 17 Sustainable Development Goals

Priority action areas

- 2) Transforming learning environments
- 3) Building capacities of educators
- 4) Empowering and mobilizing youth
- 5) Accelerating local level actions

Strategic objective

Promote ESD as a key element of quality education and a key enabler of all 17 Sustainable Development Goals with special attention to a) individual transformation, b) societal transformation & c) technological advances

Target groups

Policy-makers, institutional leaders, learners, parents, educators, youth & communities

ACTIVITIES BY MEMBER STATES: COUNTRY LEVEL MULTI-STAKEHOLDER INITIATIVE

1



Advancing policy

Integrating ESD in global, regional & national policies of education & sustainable development

2



Learning Environment

Promoting whole-institution approach

3



Educators

Providing capacity development opportunities

4



Youth

Providing opportunities for youth engagement

5



Community

Empowering local communities as “nodal” platforms for all priority action areas

Country level multi-stakeholder initiative on ESD

Normative guidelines & country-level support

Inclusive network of partners

Provide regular platforms to meet & exchange

Monitoring & evaluation

Monitor achievements in the 5 priority action areas & programme implementation, in line with SDG Target 4.7 monitoring

Communicating for action

Develop a dedicated programme to inform about the SDGs & the role of education

Evidence-informed implementation & resource mobilization

Monitor emerging issues & trends and mobilize resources to support ESD implementation

UNESCO SUPPORT TO MEMBER STATES



Community-based ESD practices and policy



Okayama Commitment 2014:Japan Case Study



- <https://www.thinkingsustainably.com/teaching-sustainability/>

- **Japan , Okayama Commitments :**

- Environmental conservation
- Disaster risk reduction (DRR)
- Income generation, entrepreneurship, community development
- Cultural diversity and dialogue, intergenerational exchange
- Literacy
- Empowerment



The Six action principles are:

- **Responding to emergent local community issues** that impact on achieving sustainable development
- **Engaging with different key players** to respond effectively to the issues identified
- **Enabling all those who seek solutions** to sustainability issues to contribute actively to finding them
- **Embedding the principles** and practices of **ESD** in daily life
- **Sustaining changes** and achievements beyond short-term project timelines
- **Transforming ourselves**, our practices and policies towards a more sustainable future

The four policy support mechanisms, :

- Providing resources
- Networking/partnership
- Capacity-building
- Content development



How Education is related to other SDG targets(UNESCO, 2016a)



Goal 1 Education is critical to lifting people out of poverty.

Goal 2 Education plays a key role in helping people move towards more sustainable farming methods, and in understanding nutrition.

Goal 3 Education can make a critical difference to a range of health issues, including early mortality, reproductive health, spread of disease, healthy lifestyles and well-being.

Goal 5 Education for women and girls is particularly important to achieve basic literacy, improve participative skills and abilities, and improve life chances.

Goal 6 Education and training increase skills and the capacity to use natural resources more sustainably and can promote hygiene.

Goal 7 Educational programmes, particularly non-formal and informal, can promote better energy conservation and uptake of renewable energy sources.

Goal 8 There is a direct link among such areas as economic vitality, entrepreneurship, job market skills and levels of education.

Goal 9 Education is necessary to develop the skills required to build more resilient infrastructure and more sustainable industrialization.

Goal 10 Where equally accessible, education makes a proven difference to social and economic inequality.

Goal 11 Education can give people the skills to participate in shaping and maintaining more sustainable cities, and to achieve resilience in disaster situations.

Goal 12 Education can make a critical difference to production patterns (e.g. with regard to the circular economy) and to consumer understanding of more sustainably produced goods and prevention of waste.

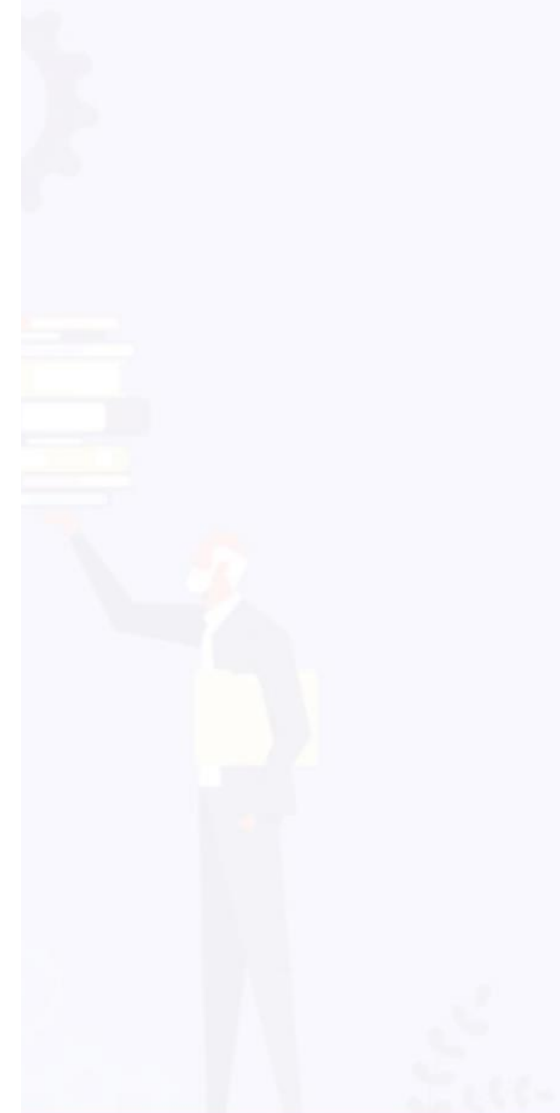
Goal 13 Education is key to mass understanding of the impact of climate change and to adaptation and mitigation, particularly at the local level.

Goal 14 Education is important in developing awareness of the marine environment and building proactive consensus regarding wise and sustainable use.

Goal 15 Education and training increase skills and capacity to underpin sustainable livelihoods and to conserve natural resources and biodiversity, particularly in threatened environments.

Goal 16 Social learning is vital to facilitate and ensure participative, inclusive and just societies, as well as social coherence.

Goal 17 Lifelong learning builds capacity to understand and promote sustainable development policies and practices.





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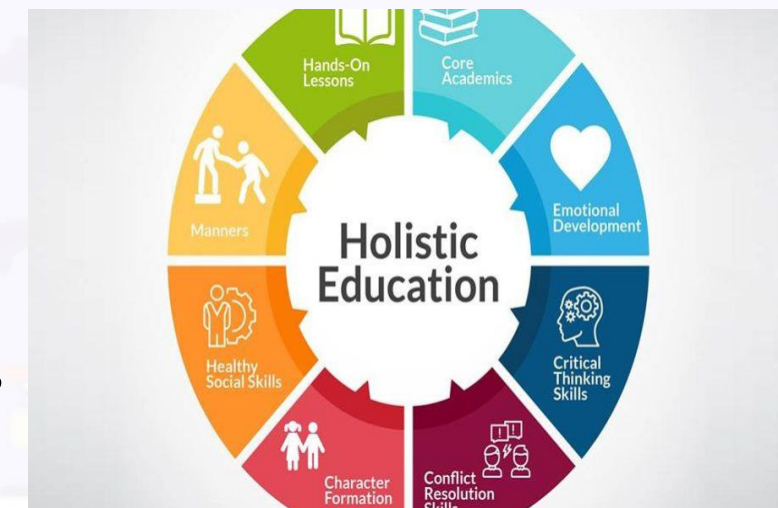
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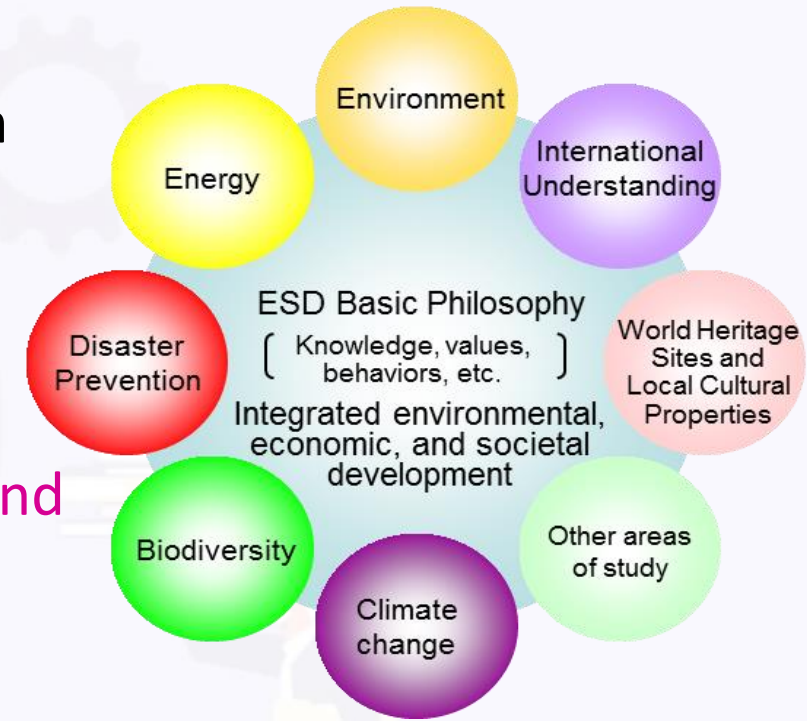




- Holistic, multidisciplinary education that integrates the social, environmental and economic pillars of sustainability and enables learners to contribute to more sustainable societies are widely expected to play a crucial role in addressing these challenges.
- Embarking on the path of sustainable development will require a profound transformation of how we think and act;
- This kind of transformation can only be accomplished through learning that fosters critical reflection on the prevailing worldviews and systems ;
- Education has long played a role in the global development agenda and has been said to lie at the heart of the Sustainable Development Goals (SDGs);
- Addressing complex and multifaceted challenges to sustainability demands innovative thinking regarding the nature and purpose of education and learning.
- Teachers are called upon to translate international policies for sustainability education into classroom practice, but research shows that many practicing educators feel unprepared to help learners develop the competencies needed to forge more sustainable paths forward.



- Education for Sustainable Development (ESD) is “holistic and transformational” and encompasses learning content and outcomes, pedagogy, and the learning environment itself”;
- Successful implementation of ESD requires a shift in focus from teaching to learning and a multidisciplinary approach that addresses the environmental, economic and social pillars of sustainability.
- ESD calls both for integration of key topics, such as climate change, disaster risk reduction, health and well-being, equity and sustainable consumption into the curriculum, and for a shift in the teacher’s role, from purveyor of knowledge to facilitator of learning that helps students develop key sustainability competencies.
- These competencies represent “the attributes individuals need for action and self-organization in various complex contexts and situations”, and they develop from an “interplay of knowledge, capacities and skills, motives and affective dispositions” as the result of experience, action, and reflection.





Systems thinking

the abilities to recognize relationships; to analyze complex systems; to deal with uncertainty

Anticipatory

the ability to understand and evaluate multiple possible futures; to deal with risks and changes

Normative

the ability to understand and reflect on the norms and values that underlie one's actions; to negotiate sustainability values, principles, goals and targets

Strategic

the abilities to collectively develop and implement innovative actions that further sustainability at the local level and further afield

Collaboration

the abilities to learn from others; to understand and respect the needs, perspectives, and actions of others; to facilitate collaborative problem solving

Critical thinking

the ability to question norms, practices and opinions; to reflect on one's own values; take a position in the sustainability discourse

Self-awareness

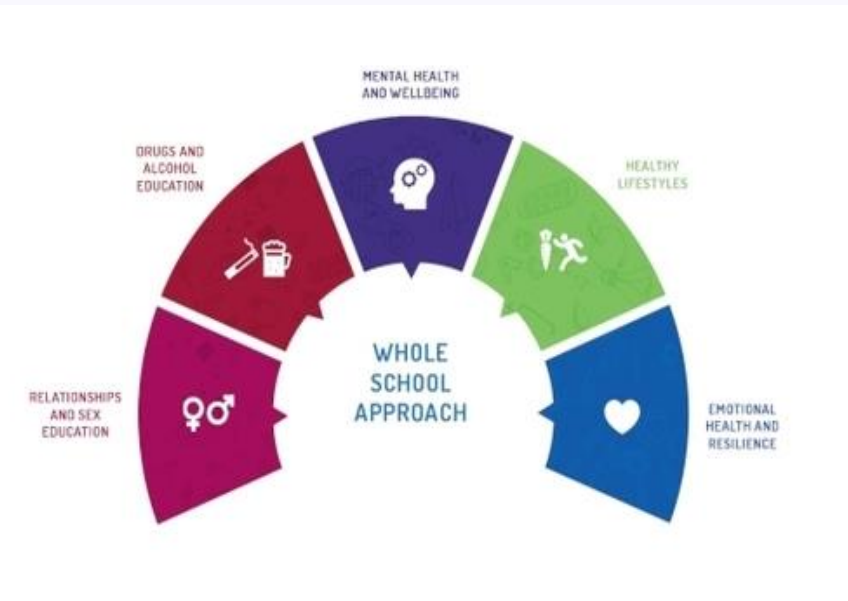
the ability to reflect on one's own role in the the local community and (global) society; to deal with one's feelings and desires

Integrated problem-solving

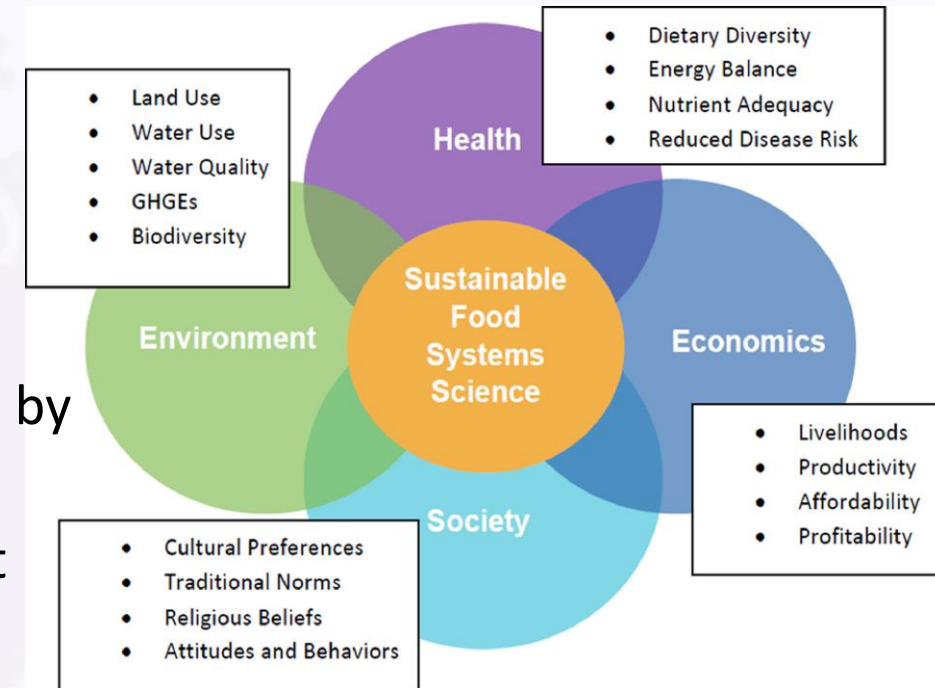
the overarching ability to apply different problem-solving frameworks to complex sustainability problems

- **Key competencies for sustainability , Source : Rieckmann M. Education for Sustainable Development Goals: Learning Objectives. Paris: UNESCO; 2017.**

- Teachers have been “singled out as the decisive factor when it comes to success in student learning” and, as translators of **ESD into classroom practice, serve as crucial change agents** who support learners in developing sustainability competencies;
- **Fostering these competencies** requires **integration of three interrelated dimensions of learning** as described by UNESCO.
- **Cognitive learning** (builds **knowledge, understanding, and critical thinking**)
- **Social and emotional learning** (develops the awareness and **social skills needed** to positively and respectfully relate to others)
- **Behavioral learning** (fosters the ability to act effectively and responsibly from local to international levels to contribute to a more sustainable world).
- Recent global survey conducted by UNESCO shows that **teachers feel more comfortable with teaching cognitive skills** than with facilitating the crucially-important **social emotional and behavioral learning needed for collaborative problem solving and taking action.**

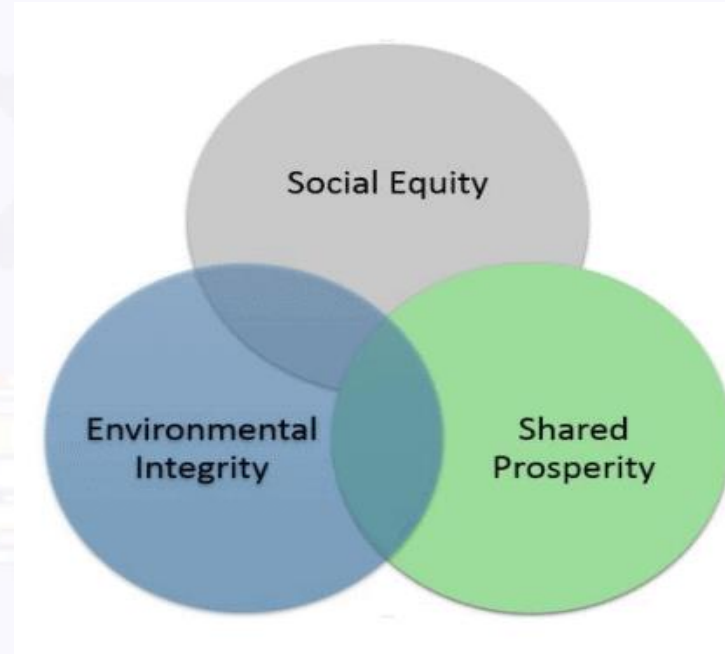


- Sustainable development means meeting the needs of all people now – including protecting the natural habitats that are essential to our survival – without compromising the ability of future generations to meet their own needs.
- Teachers help drive and embed behavior change through integrating learning about this area into teaching, both at a theoretical and practical level.
- They can also be role models in changing behavior, for example, by cycling or walking to school.
- Transformative learning has become one of the most prominent learning theory in regard to sustainable development.
- It holds enormous potential for explaining and accompanying learning processes related to processes of transformation for sustainability;
- Transformative learning can be described as a process of changing deeply held assumptions (i.e., frames of reference or meaning perspectives) about the world and oneself, thereby strengthening one's capacity to contribute to social change processes.



• The Three Pillars of Sustainability

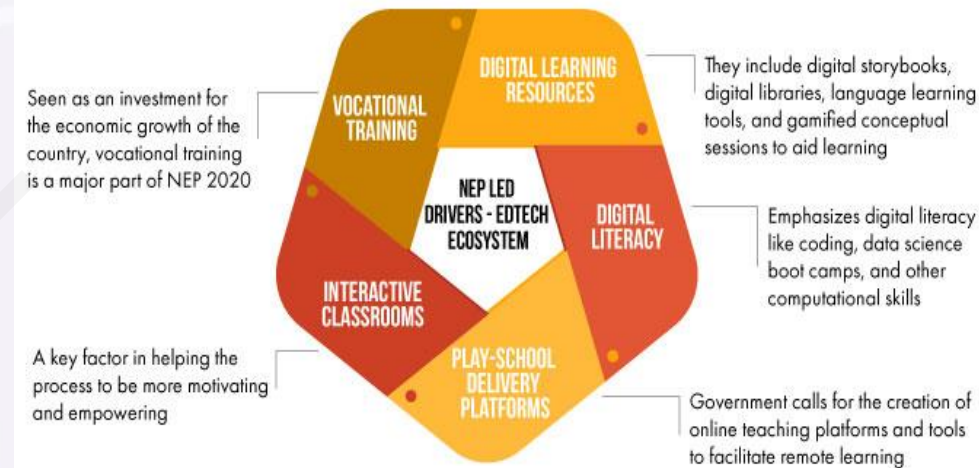
- A core belief of sustainability is that we need to consider ‘three pillars’ together: society, the economy, and the environment.
- No matter the context, the basic idea remains the same – people, habitats and economic systems are inter-related.
- If we want to have a go at achieving sustainability, we need to think about how the economic, social, and environmental aspects of any action or decision intersect.
- Focusing on only one of these issues (profit margins), can lead to environmental damages that cost society in the long run.
- But we also need a robust and healthy economy to provide the money and resources needed to take care of the environment.
- For this an educated and engaged community to help solve some of the challenges we are facing because of historical decisions.



- Access to an excellent primary education is vital for giving people the knowledge, skills, understanding, and values; they need to make decisions and take actions – all of which can shape the world.
- People who understand that our environment, economy, and society are interconnected are much more likely to make decisions and take actions that improve our quality of life without compromising the planet.
- Education for sustainability is a process that engages people and social groups in learning to live in sustainable ways.
- It encompasses a new vision of education that seeks to empower people of all ages to take responsibility for creating a sustainable future.
- Sustainability education teaches how different parts of our environment are essential for sustaining life, healthy economies, and thriving communities.



Key technology interventions of NEP 2020



Source(s): Secondary research, PGA Labs Analysis

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 Sustainability and Impact



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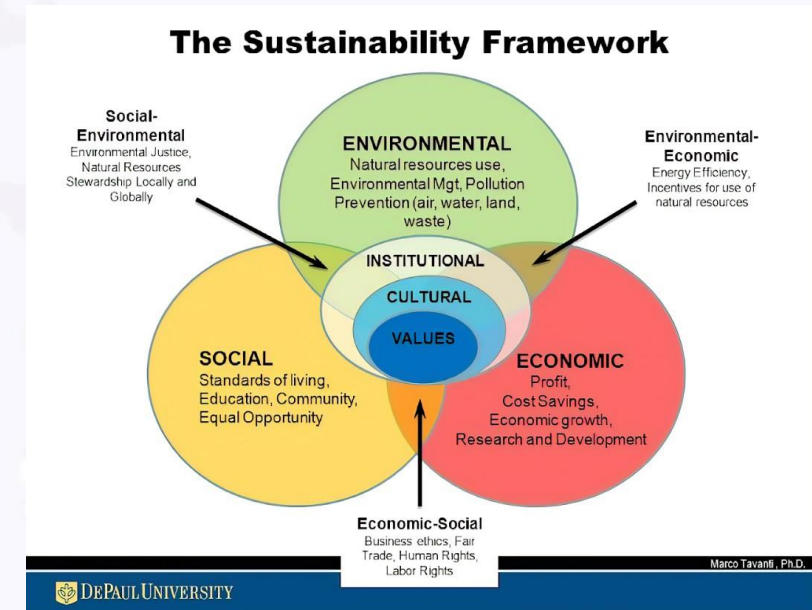


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- **The Frame-Model for Sustainability Competencies**
- Regarding the goals or objectives recommended for ESD, two major sources were identified.
- 1. There is a **need to change attitudes** among people so that they are able **to assess and address** their concerns about sustainable development.
- 2. There are goal recommendations which were mostly developed by researchers from the educational field, and which often achieve a higher degree of differentiation; analysis of existing ESD (learning) goals has revealed a significant **deficit in terms of the operationalization of the ESD output**.
- 3. Without **operationalization, and the resulting non-existent measuring instruments**, the needs for ESD and effects of ESD-related interventions (for example lessons, seminars, projects) cannot be determined empirically.

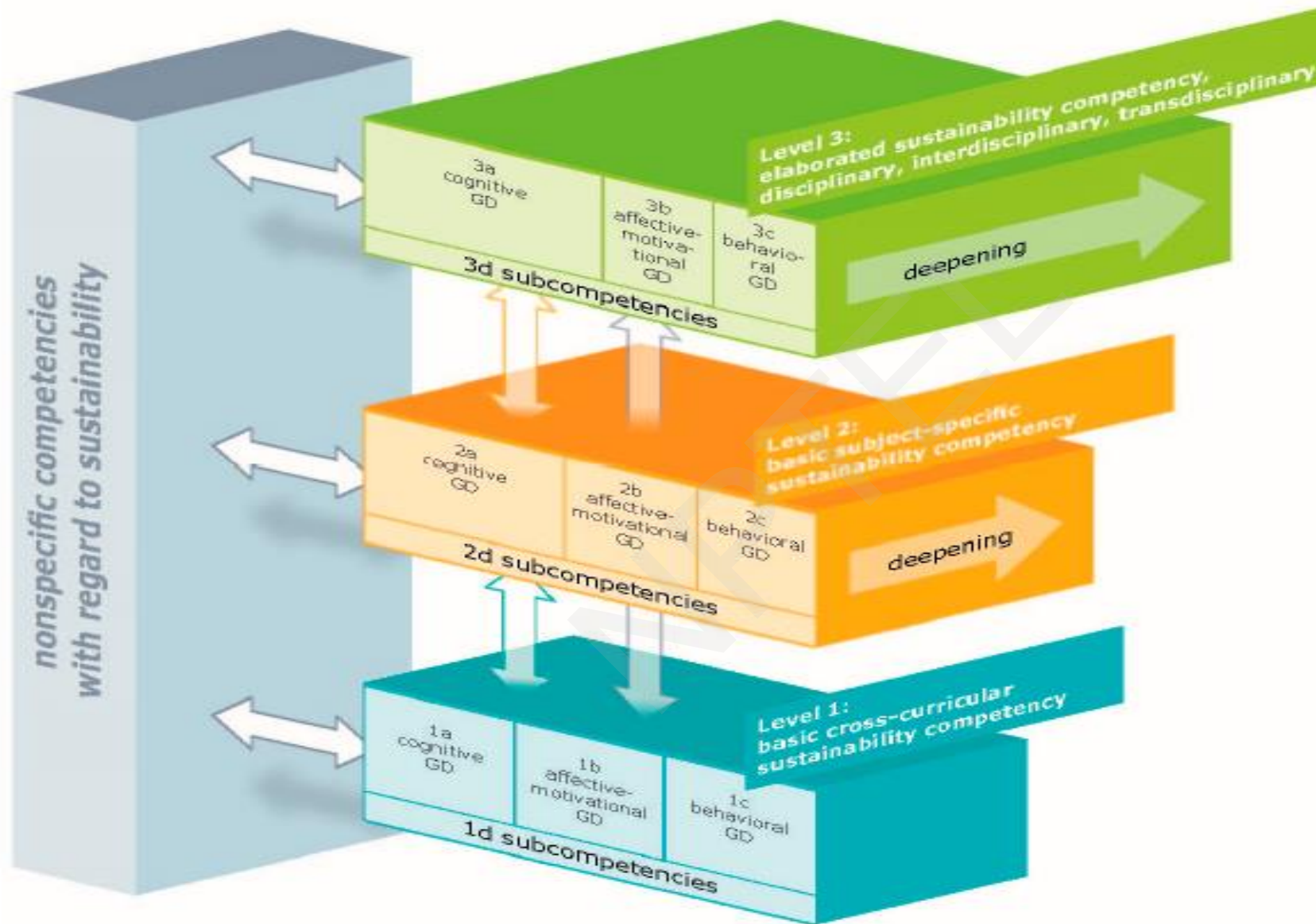


- **Learning Objectives of Sustainability**

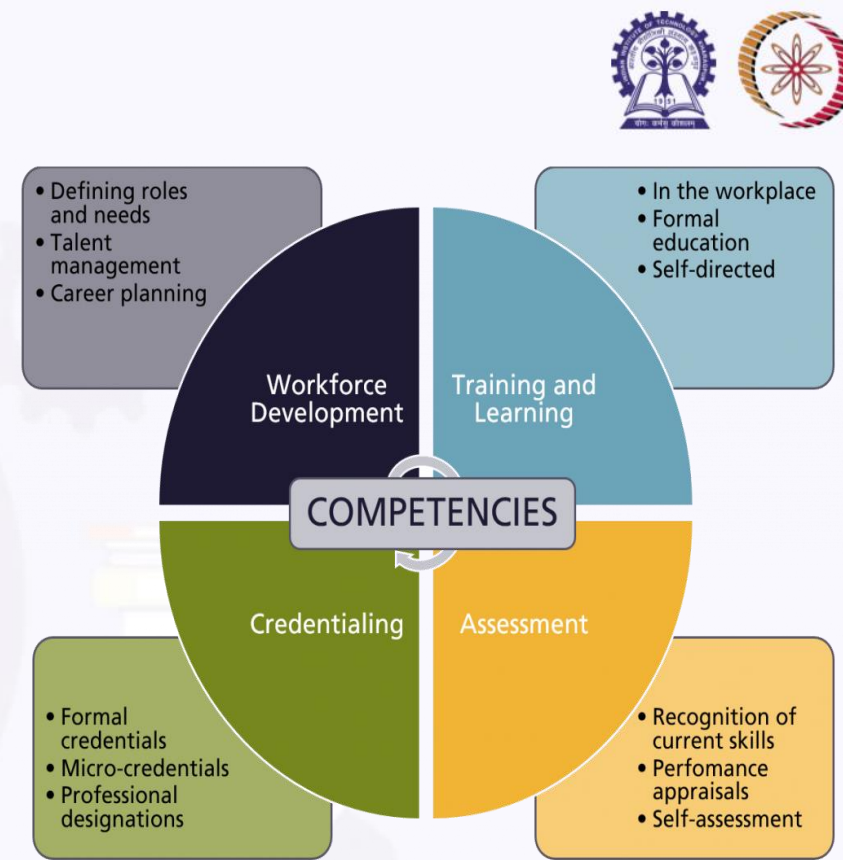
- Natural resources help drive economic growth and job creation (including as a source of food, as materials for use in manufacturing, as a place for tourism, or as an input to our medicines).
- Our environment provides other essential services that we need to survive – like clean air, clean water, soil for growing crops, climate regulation and pollination of plants.
- Time spent in nature leads to improved physical and mental health.
- We receive other intangible benefits from being in nature, such as stress-relief through meditation, the inspiration for artworks and a sense of cultural identity.
- At the same time, sustainability education teaches about the unprecedented challenges the world is facing today.
- Challenges like climate change, the alarming loss of species and habitats, increases in the number and impact of pest species, the growing risk of eco-refugees and other threats to our health, security, and future survival.
- It helps students understand that how many of the challenges we are facing are the result of individual and cumulative decisions that failed to consider sustainability effectively.
- And it helps individuals and communities develop skills to make intelligent, informed decisions about how they can help take care of the environment.



How can this shortcoming be addressed?



- **Frame-model** for the structuring of relevant sustainability competencies with cognitive, affective-motivational, and behavioral goal dimensions (GD), and sub competencies on each level . On the left, the interacting nonspecific competencies (with regard to sustainability);
- In the sense of hot cognitions ,the affective-motivational facets of sustainability competencies (Figure 1, field 1b, 2b, 3b) include all affect-, need-, and motivation-related competency features.
- These include, among other things, values (such as personal acceptance of the intergenerational idea of justice or the personally favored lifestyle), attributions of responsibility or attitudes (for example in questions of consumption or mobility (on Level 1 and 2) and mature epistemological beliefs about the relativity and situational character of empirical knowledge in important areas of sustainability on Level 3).
- The underlying affective-motivational traits become progressively more conscious on the way from Level 1 to Level 3. An affective goal commitment or a positively assessed sequence of actions is the core of any motivation, without which an action will not be performed.



- **German Research Study:**

- **Research Question 1:** What is the present state of (lower secondary) student sustainability competencies in Germany (Baden-Wuerttemberg) after the implementation of ESD as a new guiding principle?
- **Research Question 2:** What kinds of subscales can be developed and used to capture sustainability competencies?
- The measurement instrument presented here provides a valuable starting point for further test extension, e.g., for assessing learning progression, to assess the development of competencies in multilevel analysis or longitudinal studies.
- Results revealed some first insights into the interactions of different socio-demographical aspects with the different sustainability competency dimensions.
- In future studies it will be interesting to explore the role of the interactions of the different facets of sustainability competencies in more depth and compare them to other findings of relevant research in the field of for example psychology for sustainability to further explore the interconnections.



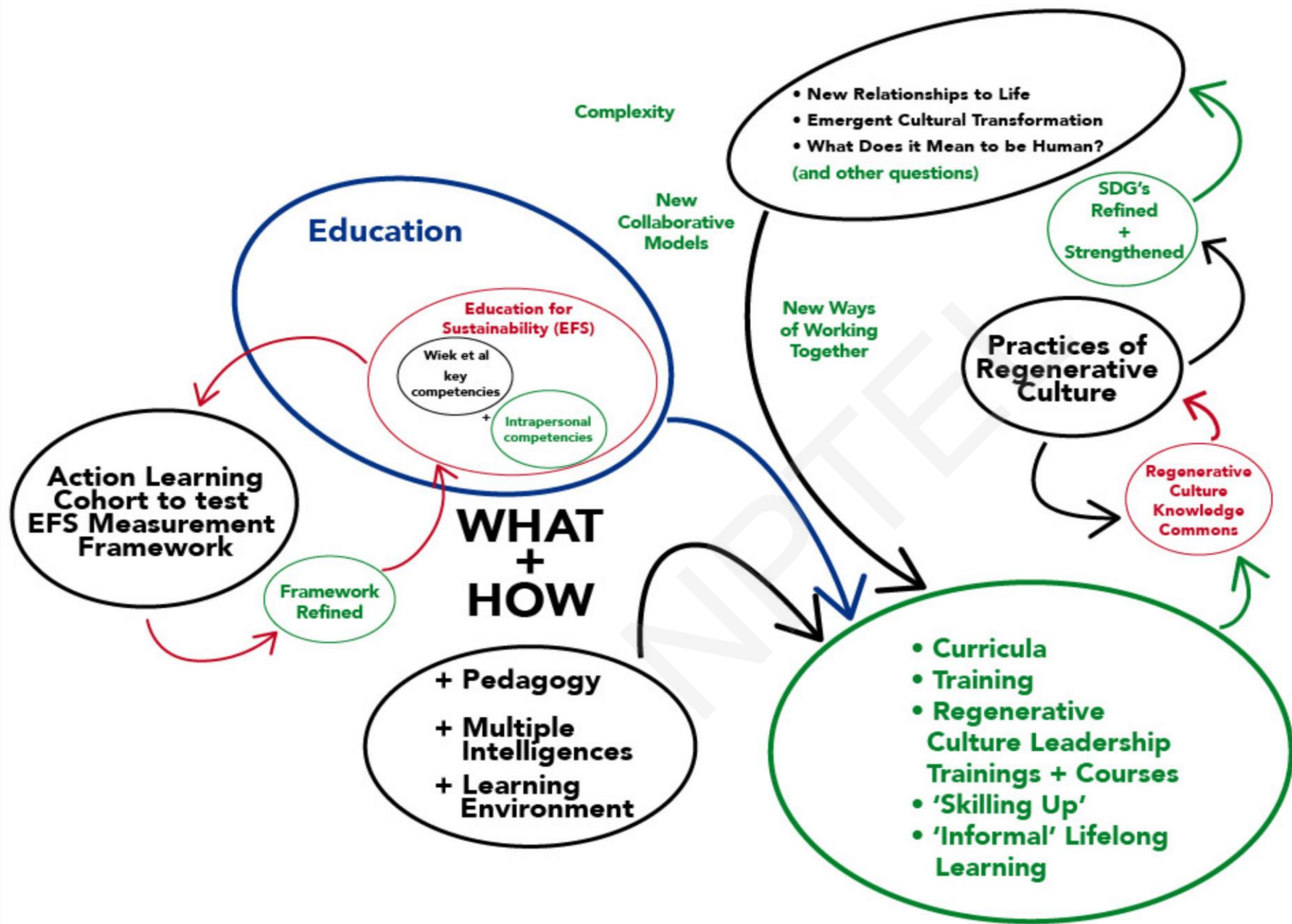
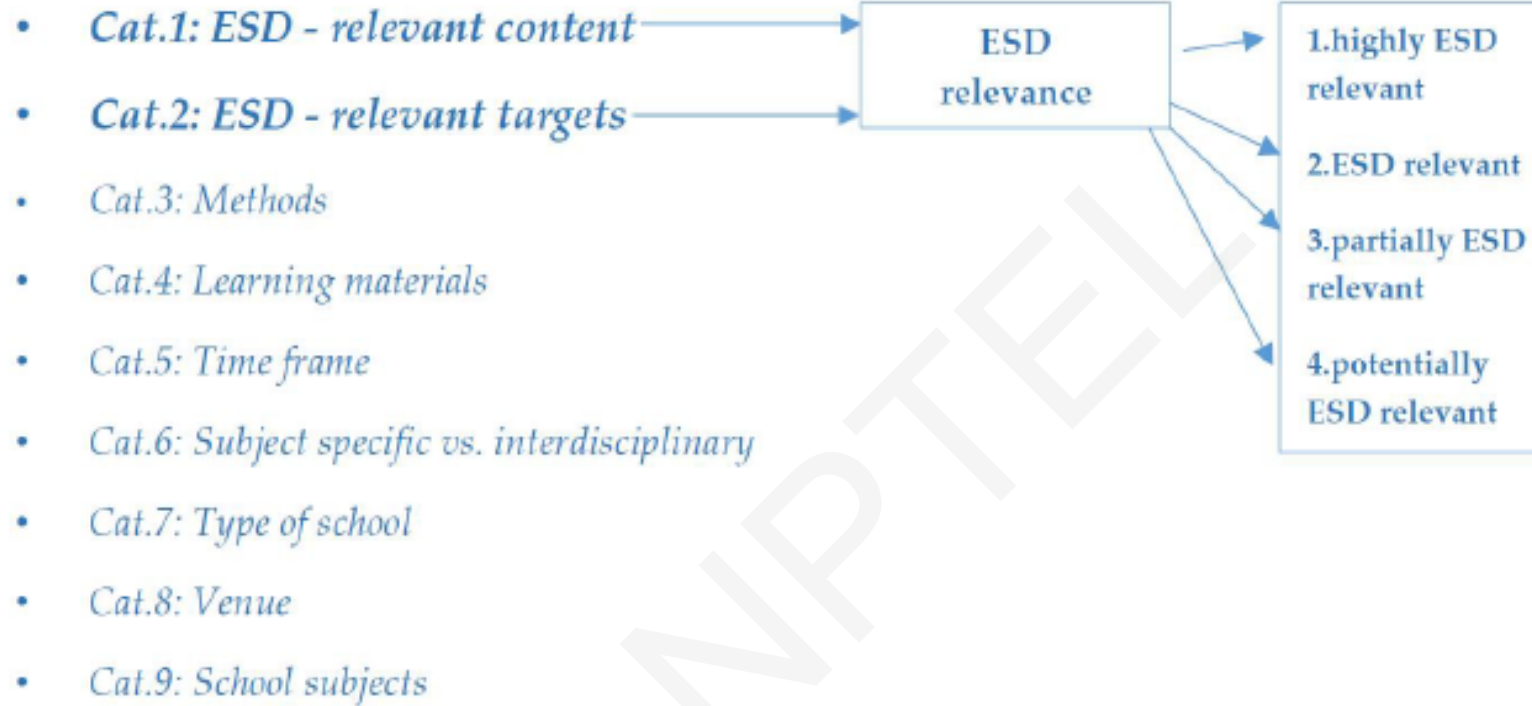


Table 1. Samples for each dimension of the questionnaire.

	Questionnaire Dimension	Sample	Item Type
1.	Socio-demographic	<i>Age, gender, chosen subject specification</i>	Mixed
2.	Basic cross-curricular and subject-specific sustainability related knowledge	<i>If someone wants to live sustainably, she/he/* should...</i> <i>Please mark only one response option (the most appropriate).</i> ☐... <i>always donate money for aid and conservation projects in poorer countries.</i> ☐... <i>if possible, eat local farming products, invest money profitably and fight for worldwide peace.</i> ☐... <i>eat as vegan as much as possible to protect local animal species, to create a fairer life for animals and humans.</i> ☐... <i>respect nature, advocate for justice, and make sure everyone has enough to live on.</i>	Single-choice items with four response options (one right answer, three distractors) ¹
3.	Affective-motivational beliefs towards sustainability	<i>When I hear of cars that consume a lot of fuel, I get angry.</i>	Ordinal scale (4-point Likert, from <i>Strongly disagree</i> to <i>Strongly agree</i>)
4.	Self-reported sustainability-related behavior intentions	<i>When I buy chocolate with my pocket money, I buy organic or fair-trade chocolate.</i>	Ordinal scale (4-point Likert, from <i>Strongly disagree</i> to <i>Strongly agree</i>)
5.	Intentional subject-specific sustainability knowledge applied class 5–6 (age group 9–14) class 7–8 (age group 11–16)	<i>recycling, smart phone production/use, decline in insect numbers</i> <i>sustainable products labels/consumption, organic agriculture, renewable energy</i>	Single-choice items with different response formats (true/false –fast response items)
6.	Dilemma situations	<i>Imagine the following situation: You are the boss of a big company and you can make all decisions for this company. Your company has earned a lot of money this year. You decide: (Please mark only one response option.)</i> ☐ <i>I install a new solar system and an electric power charging station for my employees' e-cars.</i> ☐ <i>I build a new sports area so that my staff will feel well and stay healthy.</i> ☐ <i>I hire a consultant who helps me with future decisions to make the production more environmentally friendly and improve the working conditions for the employees, and I implement this program.</i> ☐ <i>I pay myself a big salary so I have money to go on holidays with my family.</i>	Single-choice items with four response options on an ordinal ranking ²

¹ Correct answer is marked; ² The answer with the highest score covers three sustainability dimensions and is marked.

An ESD Indicator for Teacher Training & the National Monitoring for ESD: Implementation in Germany



Categories of the coding procedure for analyzing the TT course descriptions with nine categories validated and tested by external ESD experts.(Source: Sustainability 2018, 10, 2508 7 of 17)

Thematic indicator framework : Education 2030 Framework for Action

Source: UNESCO



Goal: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

1	2	3	4	5	6	7	8	9	10	11
Concept	No.	Indicator	Equity	Sex	Location	Wealth	Available	Coverage	Proposed global Indicators*	Comments
		Targets 4.1- 4.7								
4.1 By 2030, ensure that all girls and boys complete free, equitable and quality primary and secondary education leading to relevant and effective learning outcomes										
Learning	1.	Percentage of children/ young people (i) at the end of primary and (ii) at end of lower secondary achieving at least a minimum proficiency level in (a) reading and (b) mathematics	Yes	X	X	X	1-3 years		Yes	Efforts to generate global measures will focus on common concepts of minimum proficiencies drawing links between national, regional and international student assessment studies.
	2.	Administration of a nationally representative learning assessment (i) during primary (ii) at the end of primary and (iii) at the end of lower secondary education	No				1-3 years			Standards need to be developed for all assessments, national, regional and international to improve their quality. The UIS Observatory of Learning Outcomes will track national, regional and international assessments.
Completion	3.	Gross intake ratio to the last grade (primary, lower secondary)	Yes	X			Yes	c150		
	4.	Completion rate (primary, lower secondary, upper secondary)	Yes	X	X	X	Yes	c100		This indicator is currently available but work is required to finalize a common methodology and increase the number of surveys available to calculate it. The rate is calculated for young people aged 3-5 years above the official ending age for a given level of education.

- **Teacher Education for Sustainable Development in Germany :**
- **Learning objectives for teachers to promote ESD**
 - 1. Know about **sustainable development**, the different SDGs and the related topics and challenges
 - 2. Understand the **discourse on and the practice of ESD** in its local, national and global context
 - 3. Develop their own integrative view of the **issues and challenges of sustainable development** by taking into account the **social, ecological, economic and cultural dimensions** from the perspective of the principles and values of sustainable development including that of **intergenerational** and **global justice**
 - 4. Take disciplinary, **interdisciplinary and transdisciplinary perspectives** on issues of **global change** and their local manifestations





- Reflect on the concept of sustainable development, the challenges in achieving the SGDs, the importance of their own field of expertise for achieving the SDGs and their own role in this process
- Reflect on the relationship of formal, non-formal and informal learning for sustainable development, and apply this knowledge in their own professional work
- Understand how cultural diversity, gender equality, social justice, environmental protection and personal development are integral elements of ESD and how to make them a part of educational processes
- Practice an action-oriented transformative pedagogy that engages learners in participative, systemic, creative and innovative thinking and acting processes in the context of local communities and learners' daily lives
- Act as a change agent in a process of organizational learning that advances their school towards sustainable development

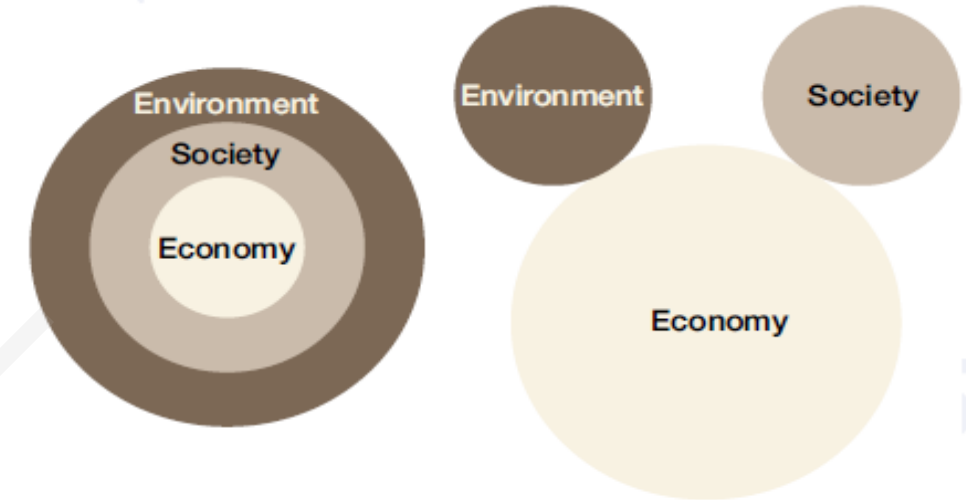




- Identify **local learning opportunities** related to sustainable development and **build cooperative relationships**
- Evaluate and assess the learners' development of **cross-cutting sustainability competencies** and specific sustainability-related **learning outcomes**
- Possible modules of a **teacher education curriculum with ESD** as a key element.
- Basic concepts of **sustainable development** from a **local, national and international perspective**
- **ESD concepts from a local, national and international perspective**
- **Disciplinary, interdisciplinary and transdisciplinary views** of key examples of sustainability challenges
- **Project-oriented work** on specific problems of local, national and global importance in **cooperation with educational institutions** and other (local) partners
- **Research-based analysis of ESD** processes in different learning settings (such as schools, colleges or non-formal educational institutions)
- **Practical experiences with ESD approaches** and their **critical reflection**



UNESCO ESD Sourcebook (UNESCO 2012), whole-school approaches



VISUALIZING SUSTAINABILITY



- The formal **curriculum contains** knowledge, skills, perspectives and values **related to sustainability**.
- **Learning** includes **real-life issues** to enhance pupils' motivation and learning.
- The **school has a sustainability ethos**, which can be seen in the treatment of others, school property, and the environment.
- **School management** practices **reflect sustainability**.
- **School policies reflect** environmental, social, and economic sustainability.
- **Interactions** between the school and community are fostered.
- Special events and extra-curricular activities apply and enhance **classroom learning about sustainability**.
- Pupils engage in **decision-making affecting school life**.



Monitoring Progress: The SDG India Index



Ranking

State/UT Performance

Methodology

Metadata

Select Goal

SDG India Index

State Scores



Performance category

	Category	States
1	Achiever	
2	Front Runner	Himachal Pradesh, Kerala, Tamil Nadu, Chandigarh, Puducherry
3	Performer	Andhra Pradesh, Arunachal Pradesh, Chhattisgarh, Goa, Gujarat, Haryana, Jammu and Kashmir, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Punjab, Rajasthan, Sikkim, Telangana,

Goal-wise ranking of states/ UTs and overall ranking based on performance on all goals

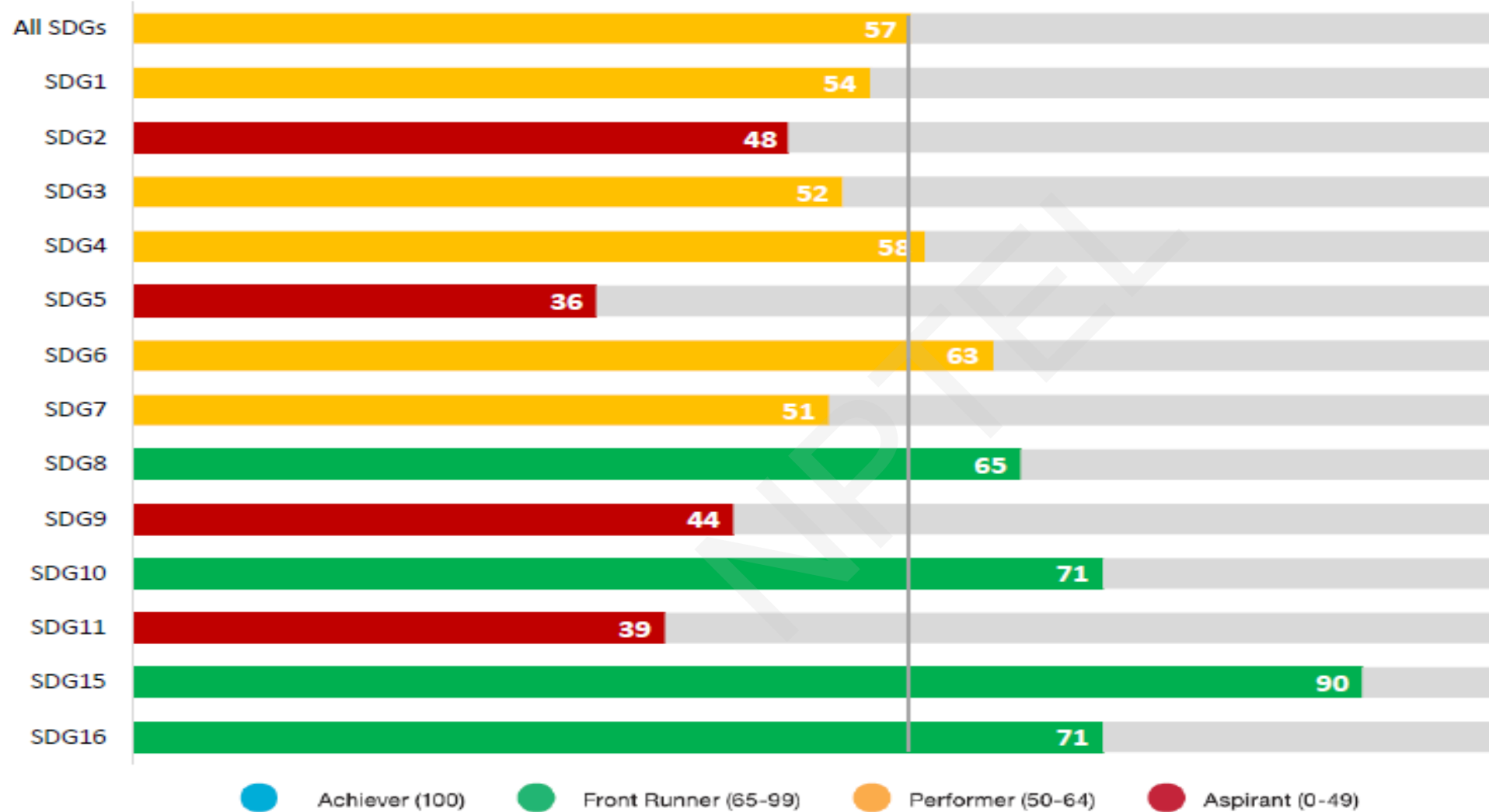
Promotes competition among the states/ UTs

Enable states/ UTs to learn from peers

Supports states/ UTs in identifying priority areas

Highlights data gaps

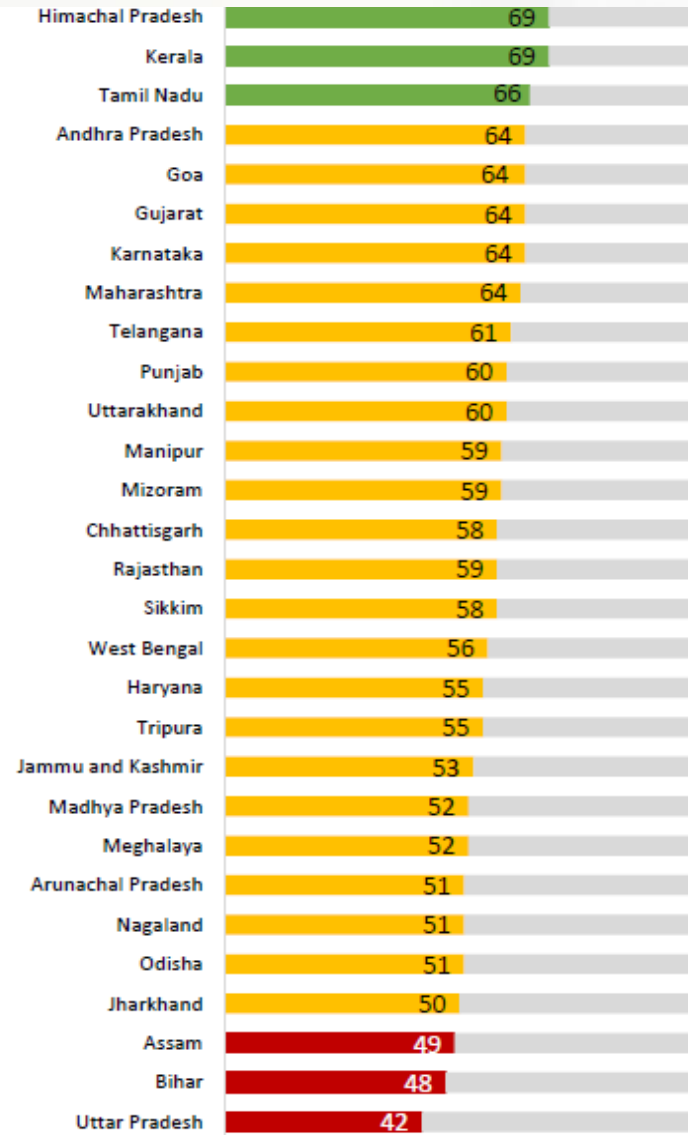
INDIA'S INDEX SCORE: OVERALL AND SDG-WISE



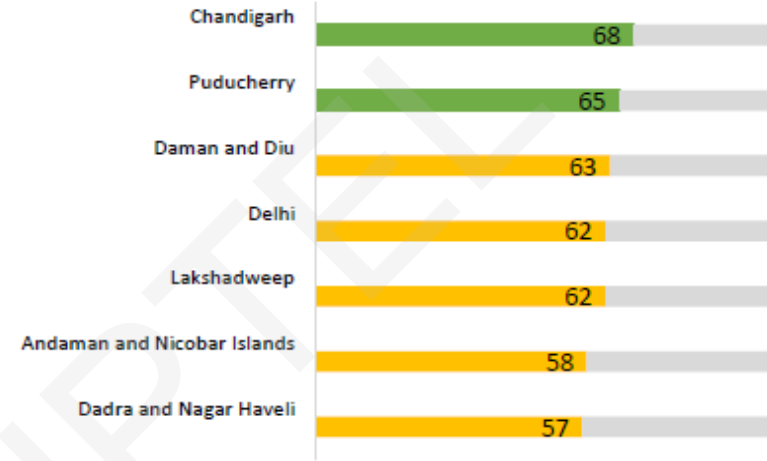


OVERALL SCORE

STATES



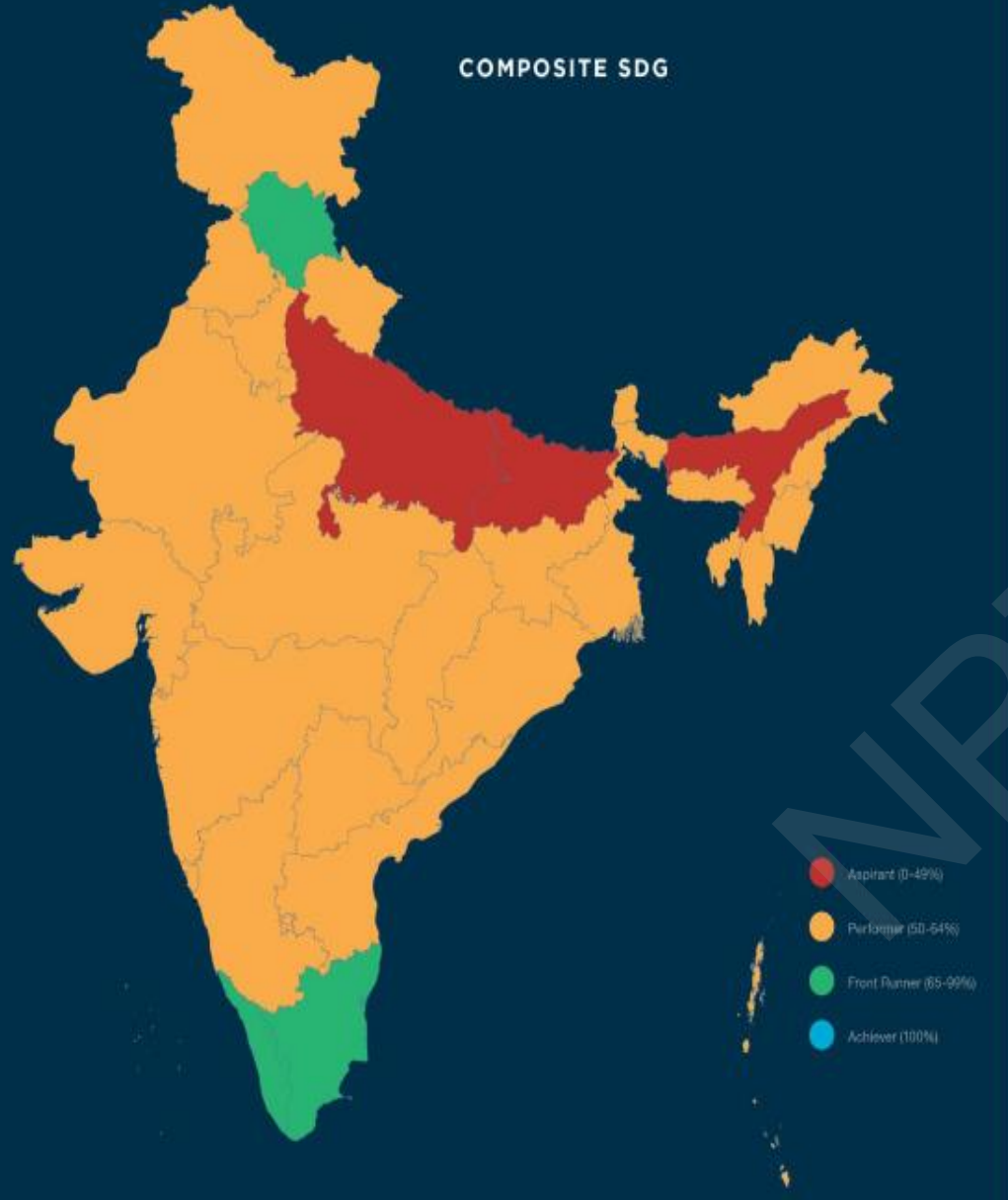
UNION TERRITORIES



● Achiever (100) ● Front Runner (65-99) ● Performer (50-64) ● Aspirant (0-49)



COMPOSITE SDG



Front Runners

Chandigarh
Himachal Pradesh
Kerala
Puducherry
Tamil Nadu

Performers

A & N Islands
Andhra Pradesh
Arunachal Pradesh
Chhattisgarh
Dadra & Nagar Haveli
Daman & Diu
Goa
Gujarat
Haryana
Jammu & Kashmir
Jharkhand
Karnataka
Lakshadweep
Madhya Pradesh
Maharashtra
Manipur
Meghalaya
Mizoram

Nagaland
Odisha
Punjab
Rajasthan
Sikkim
Telangana
Tripura
Uttarakhand
West Bengal

Aspirants

Assam
Bihar
Uttar Pradesh

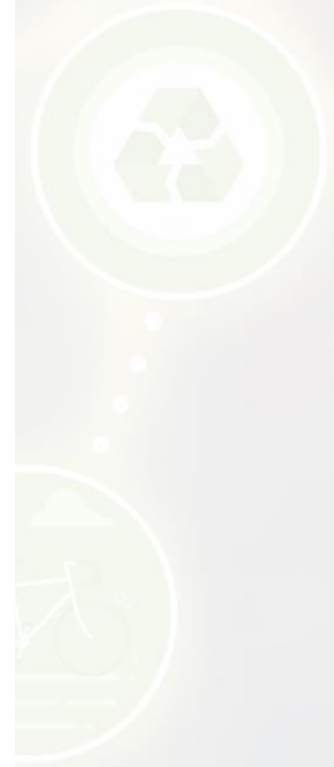
Meeting Challenges...

Generating data
on all SDG
Indicators from
local to national
level

Convergence and
efficiency of
financing SDGs
across various
actors and
institutions

Capacity
development of
multiple
stakeholders on a
range of gaps

Identifying
regional focus
areas &
developing
action strategies



Emerging Focus Areas



SDG Financing

- Mobilizing **public finance**: Gap analysis followed by interventions in tax capacity, enhancing spending efficiency and rationalization of spending, assessing fiscal sustainability and space



Engaging the Private Sector

- Mainstreaming SDGs in core business beyond the CSR domain
- Investment in **research and innovation** in new technology and low-cost solutions in areas like waste management, resource efficiency etc.



Strengthening the Data System

- Operationalising the **State and District Indicator Frameworks**
- Functional systems for ensuring data integrity



Leveraging Science, Technology, Innovation

- Using STI to **Leave No One Behind** & deliver quality public services for everyone
- STI platforms to promote **entrepreneurship** & business development





**UNESCO Futures of Education Report:
Reimagining our futures together**



<https://en.unesco.org/news/united-nations-alert-education-should-be-clear-priority>

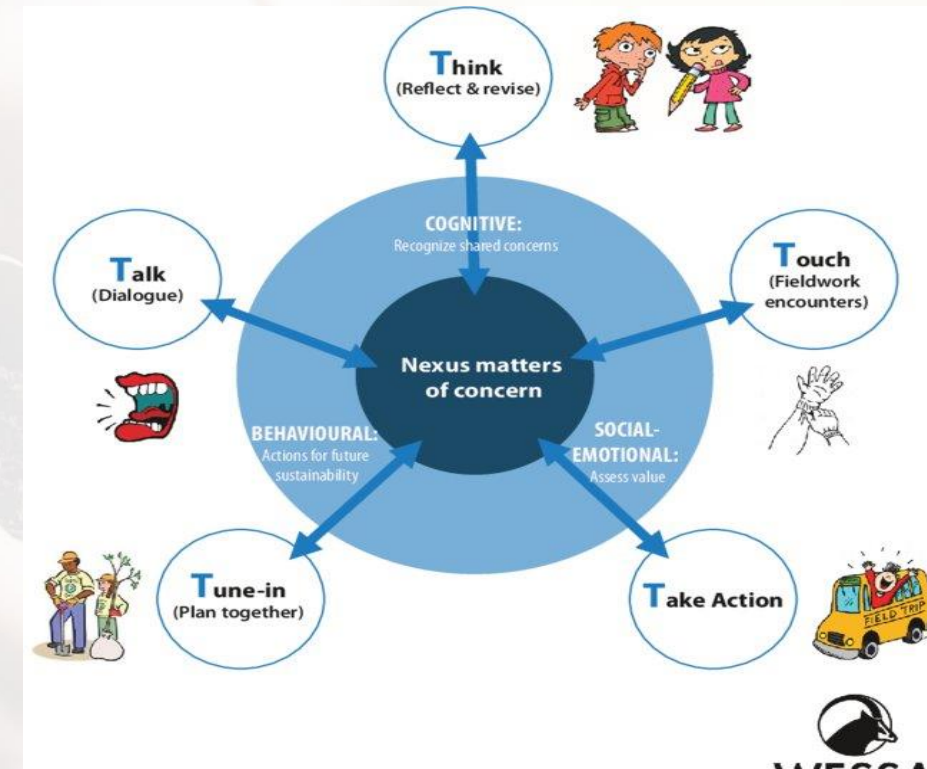
<https://www.youtube.com/watch?v=j8aXSTN71MY>

<https://en.unesco.org/news/sdg4youth-network-positioning-students-and-youth-front-and-centre-sdg4>

<https://en.unesco.org/news/global-media-and-information-literacy-week-media-and-information-literacy-public-good>



- **Perspectives of Sustainable Development**
- 1. To achieve sustainable development and a **higher quality of life for all** people, countries should reduce and **eliminate unsustainable patterns** of production and **consumption** and **promote appropriate** demographic **policies**.
- 2. **Women play a vital role** in environmental management and development. Their **full participation** is therefore essential to achieving sustainable development.
- 3. **Warfare is inherently destructive** to sustainable development. Peace, development and environmental protection are interdependent and indivisible.
- 4. **A systems thinking approach**, rather than an approach that looks at problems in isolation should be used. **Sustainability issues are linked and part of a “whole.”**
- 5. **Understanding local issues in a global context** and recognizing that solutions to local problems **can have global consequences**.





- 6. Realizing that individual **consumer decisions affect** and give rise to **resource extraction** and manufacturing in distant places.
- 7. **Considering differing views** before reaching a decision or judgement.
- 8. Recognizing that **economic values, religious values, and societal values compete** for importance as people with different interests and backgrounds interact.
- 9. Seeing all humans as having **universal attributes**.
- 10. Knowing that **technology and science alone cannot solve all** of our problems.
- 11. Emphasizing the role of **public participation in community** and governmental decision-making. **People** whose lives will be affected by **decisions must be involved** in the process leading to the decisions.
- 12. Calling for **greater transparency and accountability** in governmental decision-making.

Key characteristics of ESD



- **Interdisciplinary and holistic**
- **Values-driven**
- **Focused on critical thinking and problem solving**
- **Multi-methodological**
- **Participatory in decision-making**
- **Locally relevant**

- **12. Employing the precautionary principle** – taking action to avoid the possibility of serious or irreversible environmental or social harm even when scientific knowledge is incomplete or inconclusive.
- Therefore, **educators, leaders, and citizens recognize** that sustainable development is an evolving concept and that the list of sustainability **perspectives** can therefore grow and change.
- Governments and civil society as well as individuals must hold the **responsibility for a more sustainable future**. All must contribute in their own way.
- **Global Action Programme (GAP) on ESD states that:**
- ESD allows every human being to acquire the knowledge, skills, values and attitudes that empower them to contribute to sustainable development and take informed decisions and responsible actions for environmental integrity, economic viability, and a just society for present and future generations. ESD promotes skills like critical thinking, understanding complex systems, imagining future scenarios, and making decisions in a participatory and collaborative way (UNESCO, 2014b: 33).



Development of sustainability competencies:

The GAP highlights learning outcomes that stimulate learning and promote core competencies, such as 'critical and systemic thinking, collaborative decision-making, and taking responsibility for present and future generations' (UNESCO, 2014b: 12).

Key competencies for sustainable development

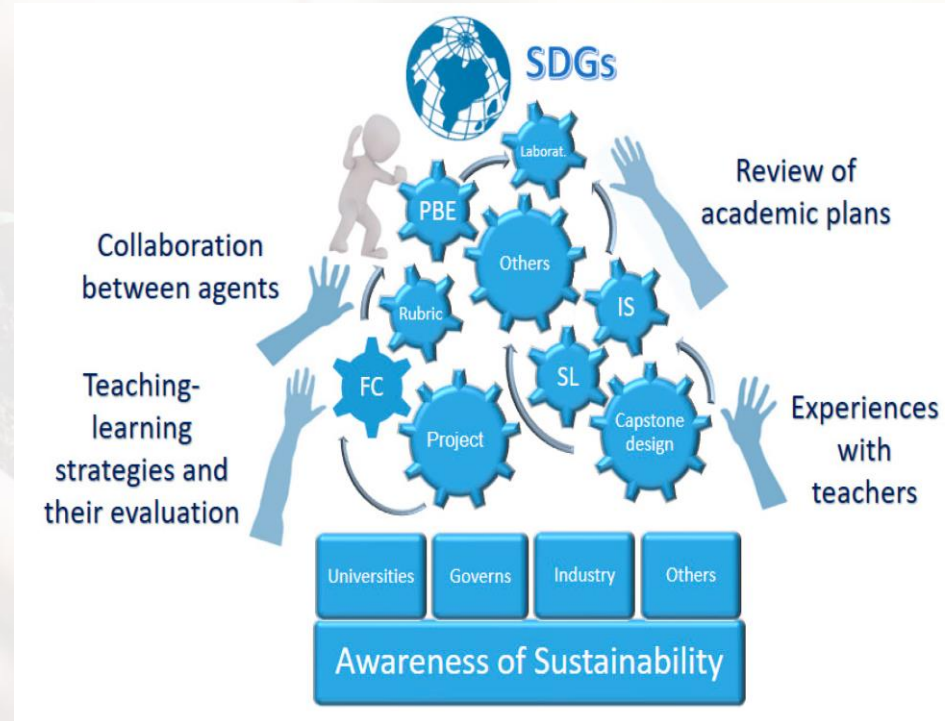
(Wiek et al., 2016):

- **Anticipatory competency:** the ability to understand and evaluate

multiple futures – possible, probable and desirable – and to create

one's own visions for the future, to apply the precautionary principle, to assess the consequences of actions, and to deal with risks and changes;

- **Normative competency:** the ability to understand and reflect on the norms and values that underlie one's actions and to negotiate sustainability values, principles, goals and targets, in a context of conflicts of interests and trade-offs, uncertain knowledge and contradictions;



- **Strategic competency:** the ability to collectively develop and implement innovative actions that further sustainability at the local level and further a field;
- **Collaboration competency:** the ability to learn from others; understand and respect the needs, perspectives and actions of others (empathy); understand, relate to and be sensitive to others (empathic leadership), deal with conflicts in a group; and facilitate collaborative and participatory problem-solving;
- **Critical thinking competency:** the ability to question norms, practices and opinions; reflect on own one's values, perceptions and actions; and take a position in the sustainability discourse;
- **Self-awareness competency:** the ability to reflect on one's own role in the local community and (global) society, continually evaluate and further motivate one's actions, and deal with one's feelings and desires;
- **Integrated problem-solving competency:** the overarching ability to apply different problem-solving frameworks to complex sustainability problems and develop viable, inclusive and equitable solution that promote sustainable development – integrating the above-mentioned competencies.



Key competencies and performance of sustainability citizens

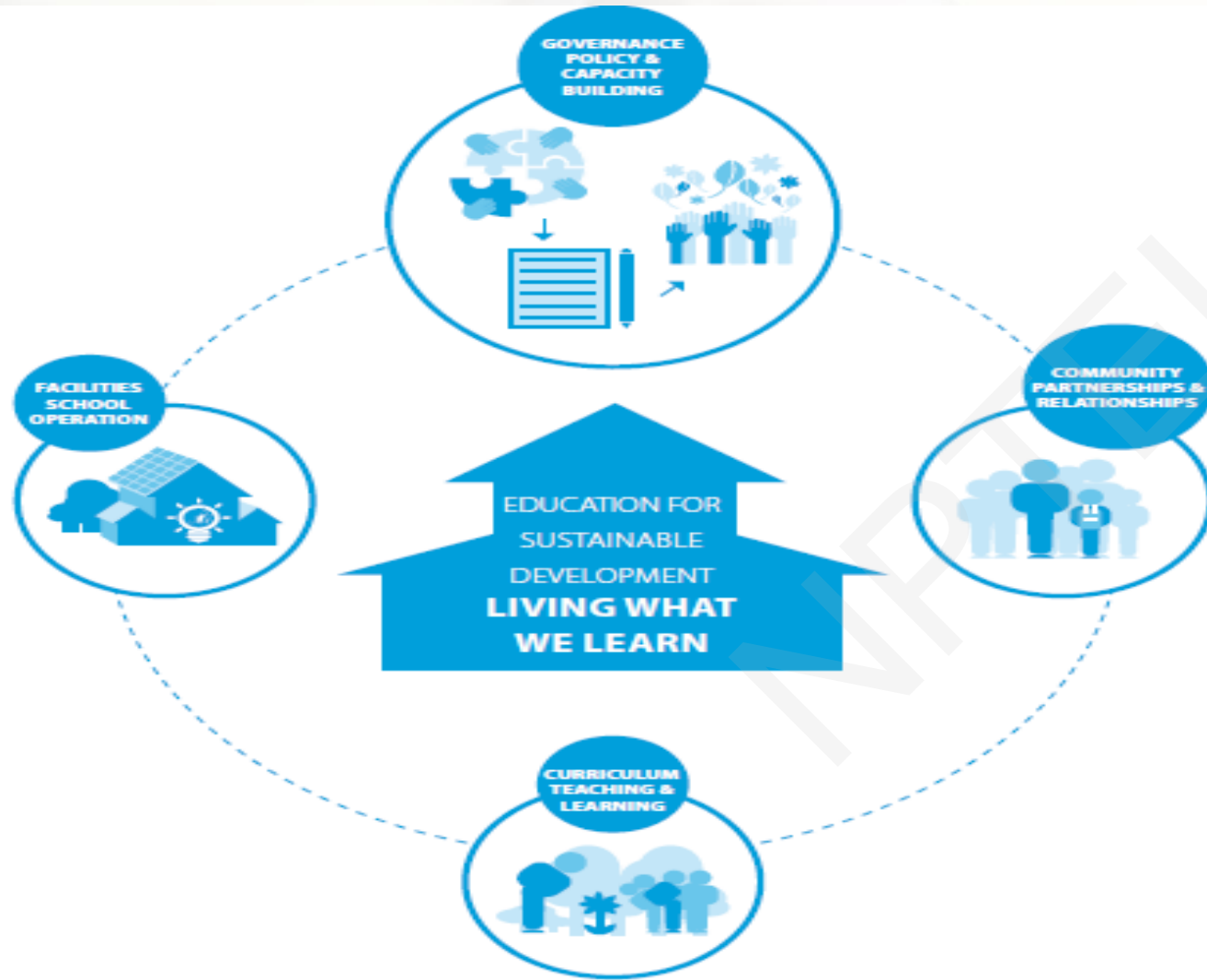
Source: (A. Leicht, J. Heiss and W. J. Byun (eds) © UNESCO 2018)



Implications of ESD for the practice of education and pedagogy



Whole-institution approach



Source: UNESCO (2014a: 89).



We are increasingly asking if what people learn is truly relevant to their lives, if what they learn helps to ensure the survival of our planet. Education for Sustainable Development can provide the knowledge, awareness and action that empower people to transform themselves and transform societies.

- Stefania Giannini, Assistant Director-General for Education, UNESCO





ESD for 2030 will promote the integration of sustainable development and the SDGs into education and learning, as well as ensuring the integration of education and learning into all activities that promote sustainable development and the SDGs



Education for Sustainable Development aims to raise knowledge, awareness and action



THANK YOU



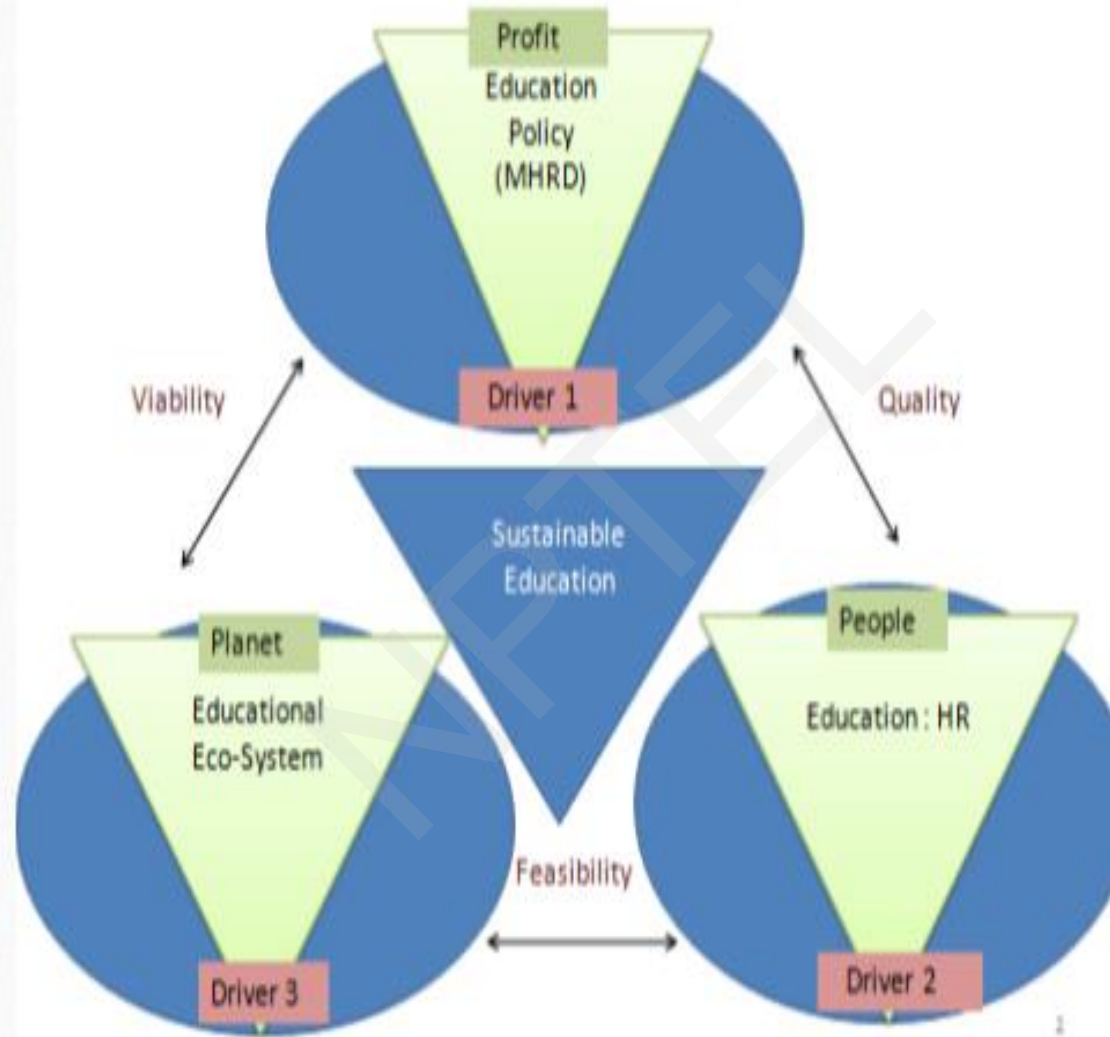
Lecture : 60

LEARNING FOR SUSTAINABILITY

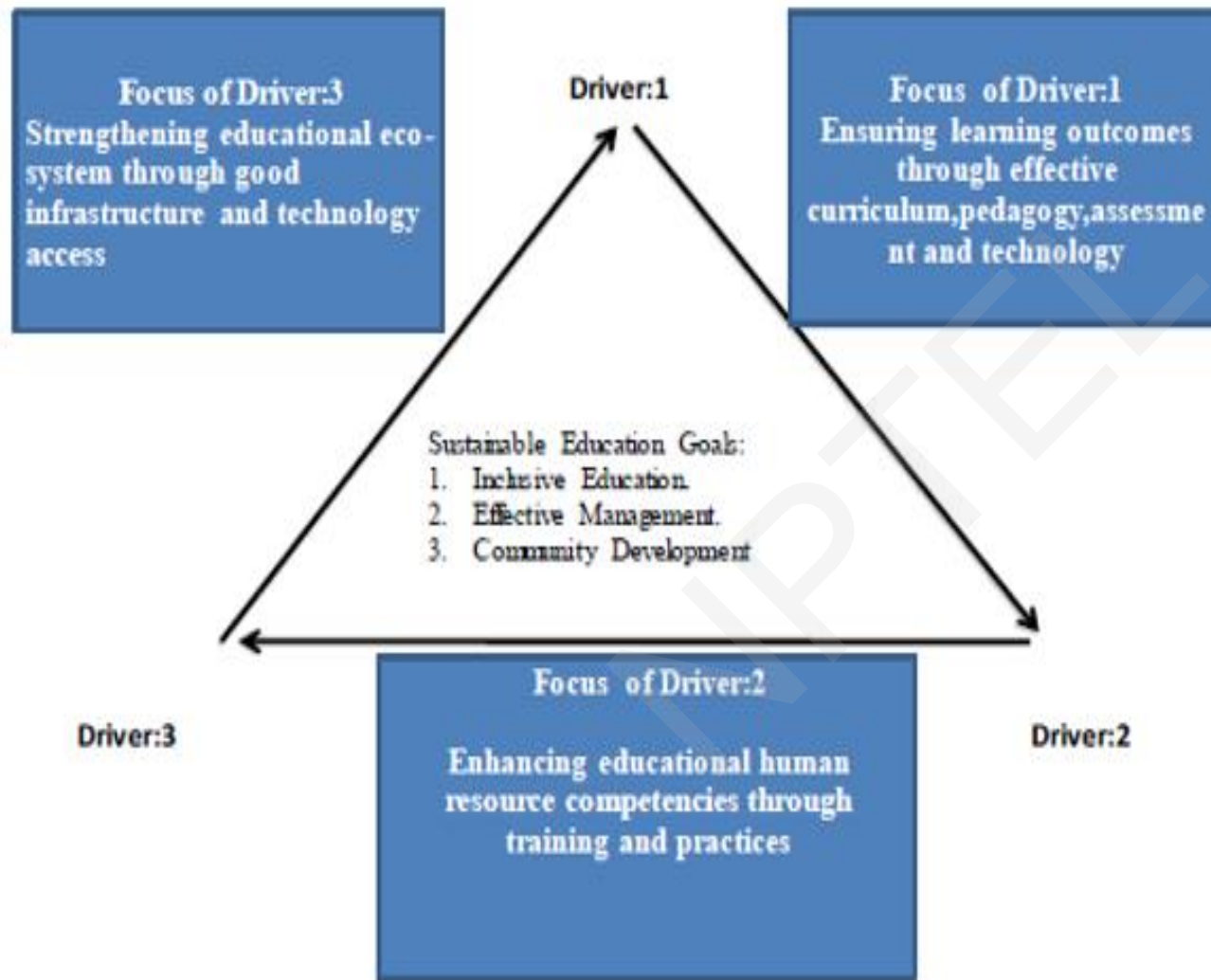
***Prof. Atasi Mohanty
IIT Kharagpur***

- Developing a Conceptual Model of Transformative Learning for the Indian Education system-

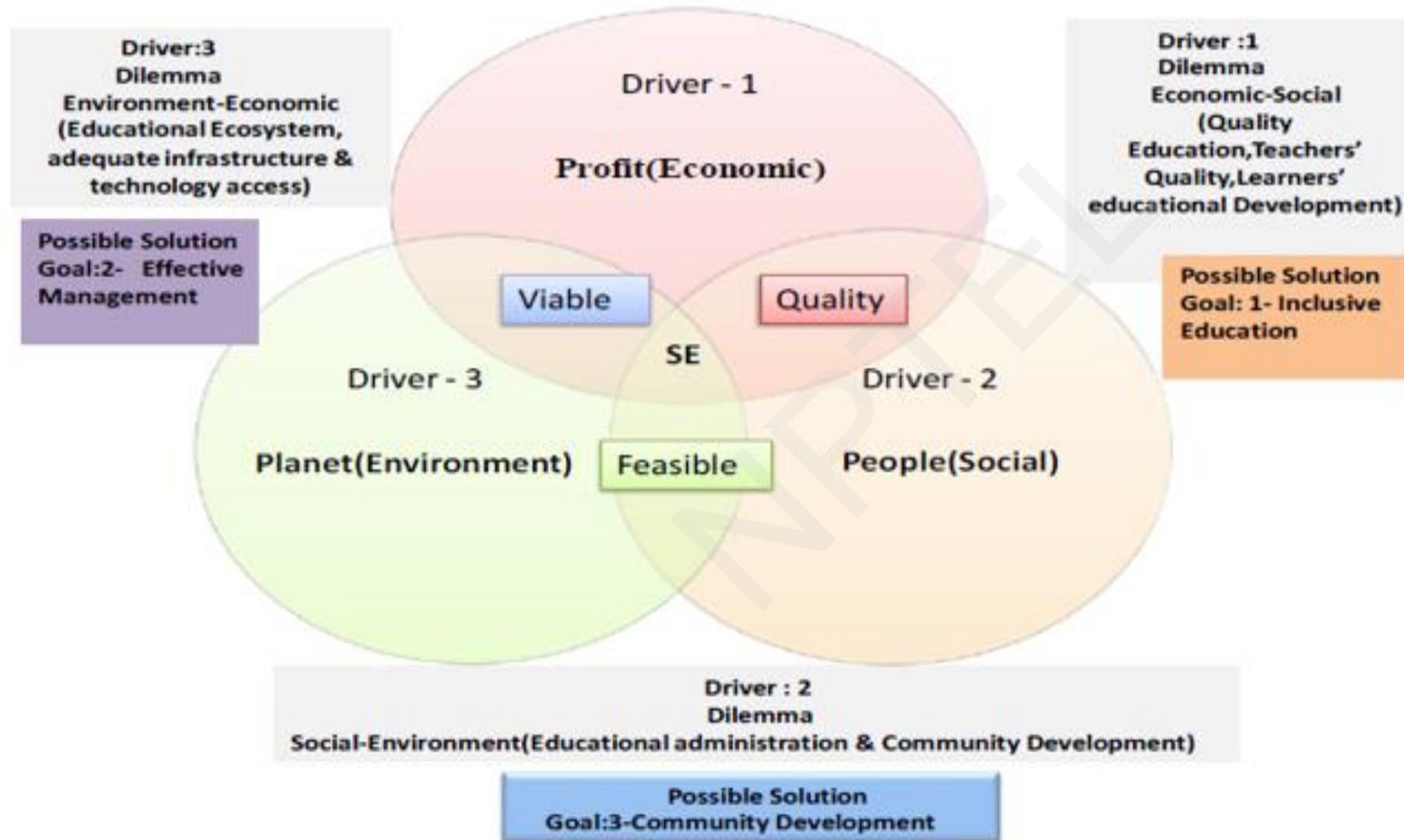
- A Conceptual Model of Sustainable Education for India



A conceptual sustainable education triangle-



A conceptual dilemma model of sustainable education-

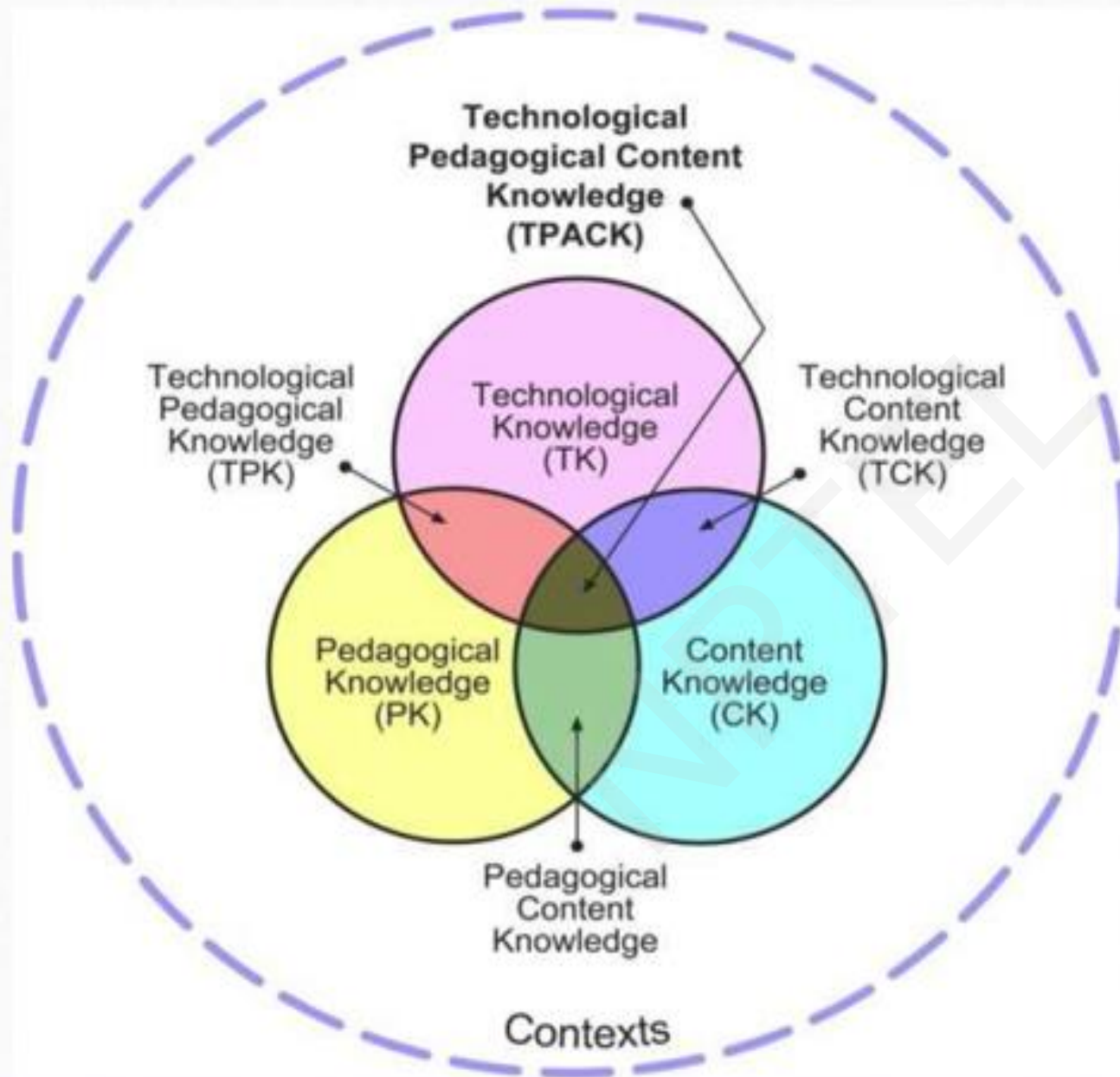


- According to (Mohanty,2019) ‘Sustainable Education’ model the vital components are, i.e., the three drivers – Driver1- Profit, Driver 2 – people, Driver 3 – the planet would be like- the 1st component **Driver – 1** is the ‘Ministry of Human Resource Development’ (MHRD) of Govt. of India as the major **policymaker, decision-taking body** and **economic resource** provider for running a centralized board of education in the country; hence called the ‘**Profit**’/Prosperity.
- The **2nd Driver is ‘people’** engaged in the education sector, the human resources (**all direct and indirect stakeholders**) of our school education system (primary or secondary level of school education);
- the **3rd Driver is ‘planet’** the **education set up, school** environment or can be called the “**educational ecosystem**” to carry on all the teaching – memorizing activities and educational administrations / transactions.





- All these three drivers (**profit, people and planet**) are expected to work together harmoniously to make the education system a sustainable one.
- Thus, the **objective of Driver 1 (profit)** would be – “Ensuring learning outcomes through effective curriculum, pedagogy, assessment and technology”.
- The **objective/ focus of Driver 2 – (people)** would be – “Enhancing educational human resource competencies through training and practices”.
- The **objective/ focus of Driver 3 – (planet)** would be – “Strengthening the educational eco-system through good infrastructure and technology access.
- New challenges to learning are emerging and **21st century education** must address to these and contribute to greater humanity in a rapidly changing world (UNESCO, 2015).
- Education has been called upon to support this **transformation** (WBGU 2011).
- The 2005 to 2014 ‘U.N. Decade of Education for Sustainable Development’ (DESD) confirmed that ESD would enable us “to constructively and creatively address present and future global challenges and create more sustainable and resilient societies ”(UNESCO 2017).
- **ESD needs to be built on transformative learning and explicitly inculcate critical reflection on values.**
- At the same time, **transformative learning** should not be used to regulate learners but **to empower them for autonomous critical action.**



• Conclusion

- In the recent years, sustainability has become a slogan for raising social, economic and environmental awareness.
- **Higher Education Institutions (HEIs) play an important role** in promoting development principles and practices.
- In addition to educating students on sustainability topics, these institutions should take on a leadership role in incorporating sustainable practices into their services and operations and in transforming their local and regional communities into more sustainable communities.
- **Acknowledging the urgent need for Sustainable development and the importance of research in this process, universities and their researchers bear the fundamental and moral responsibility to contribute with their research to Sustainable development.**



• The Role of Institutions

- The presence of a **new international agreement** or a new legislation alone is not enough to guarantee action.
- The **SDGs' ambitious vision must be converted into specific action plans** and must find a foothold in existing institutions of governance to have impact.
- It takes time for global and regional institutions to change;
- it takes longer for all **national policy priorities, legislations, planning** cycle institutional arrangements, programs, and modes of working to be dovetailed with the new paradigm, and for human, financial, and technical capacities to be put in place.
- For implementation to happen in an effective manner, **governments must invest resources and time for:**

ETH Zurich is committed to providing students and staff with skills and opportunities to approach and effectively address complex sustainable development issues



- **Awareness raising**, both among citizens at large, but also more critically among those tasked with the Agenda's implementation.
- The need for **strengthening awareness of the specifics of the Agenda among policy makers**, senior officials, and front-line workers responsible for policy implementation.
- Developing the **ability to exercise the right to education through modifying existing national policies, legislations**, and plans in line with new SDG commitments, allotting funds necessary for their implementation, strengthening implementation mechanisms, and strengthening monitoring systems.
- Strengthening their **ability to enforce the right to education** through strengthening existing redress and state accountability mechanisms.
- Building a wider **community of SDG supporters that can push for the Agenda's implementation** by creating enabling environments for civil society and citizen participation in the governance processes.



• Developing Discourse: A Way Forward

- In a challenging environment marked by, among other issues, climate change and environmental degradation, the **Global economy is undergoing significant transformations.**
- The **green transition and the move towards sustainability**, including the objective of achieving climate neutrality, are affecting how people live and work and changing skills requirements.
- Against this background, **action is needed in the education and training sector to support the green transition** and develop the sustainability competences of learners.
- Although there is increasing activity in education and training on the environment and sustainability, **learning for sustainability** is not yet a **systematic feature of education policy and practice** across the countries.
- The UNESCO is seeking to support **Member countries in their efforts to integrate sustainability into curricula, educational practice and professional development** of educators and to ensure that learners of all ages are equipped with the **knowledge to live more sustainably, develop a sense of agency** and **acquire the skills** increasingly needed in the labour market.

Typology of Green Skills



- Emphasis is also placed on **greening the activities** and operations of the education and training systems.
- The UNESCO has highlighted the key role of education and training in delivering a more **sustainable economy** and **society** and providing people with the skills they need to participate in the green transition.
- **Learning for environmental sustainability** should be **mainstreamed across educational curricula** with a lifelong learning perspective.
- Active support should be given to **teachers and trainers** to ensure they are prepared and **upskilled for the digital and green transformation** of schools and education institutions.
- In 2021, the European Centre for the Development of Vocational Training published a skills forecast scenario on the likely employment and skills implications of the European **Green Deal/EGD**.
- EGD also advocates for **Green Competences**.

Technologies for Sustainable Development



- **Green-Comp** is 'A sustainability competence that empowers learners to embody sustainability values, and embrace complex systems, in order to take or request action that restores and maintains eco-system health and enhances justice, generating visions for sustainable futures.'
- **Green-Comp** identifies four competence areas; sustainability as a competence encompasses all four areas together.
- Each area covers three competences that are sub-elements of the sustainability competence.
- For instance, the 'embracing complexity in sustainability' competence area, which includes the 'systems thinking', 'critical thinking' and 'problem framing' competences, is about empowering learners to better see the connections between specific issues and environmental changes.
- This in turn can help to correctly map challenges as a sustainability problem and take action.
- As another competence area, 'acting for sustainability', which includes the 'policy agency', 'collective action' and 'individual initiative' competences, encourages learners to act individually and collectively to shape sustainable futures, and to demand action from those responsible to tackle sustainability problems.



**Green Competences
For a Just Transition**

22 - 24 March 2022

etui.



• Taxonomy of skills for the green transition

- In 2022, the [European Commission](#) published a taxonomy(classification system) of skills for the green transition in the multilingual classification of [Skills, Competences, Qualifications and Occupations \(ESCO\)](#).
- The classification, including [381 skills](#), [185 knowledge concepts](#), and [five transversal skills](#), contributes to a common understanding of what skills are needed to meet the labour market requirements in the context of a successful and fair green transition.
- ESCO also specifies what skills and knowledge concepts are essential or optional for specific occupations.
- Examples of skills labelled as '[green](#)' include [how to conduct energy audits](#) and [measure the sustainability](#) of tourism activities, while '[green knowledge](#)' concepts include, for instance, emission standards and ecological principles. '[Green transversal skills](#)' cover, among other things, the evaluation of how personal behavior affects the environment.

'Green skills and knowledge concepts: the knowledge, abilities, values and attitudes needed to live in, develop and support a society which reduces the impact of human activity on the environment (European Centre for the Development of Vocational Training, Cedefop).

Transversal skills: Learned and proven abilities which are commonly seen as necessary or valuable for effective action in virtually any kind of work, learning or life activity. They are "transversal" because they are not exclusively related to any particular context (job, occupation, academic discipline, occupational sector, group of occupational sectors, etc.) (European Skills/Competences, Qualifications and Occupations Classification, ESCO)'.



• References

- <https://www.youtube.com/watch?v=j8aXSTN71MY>
- <https://en.unesco.org/news/global-media-and-information-literacy-week-media-and-information-literacy-public-good>
- <file:///C:/Users/Prof.%20Atasi%20Mohanty/Desktop/ESD%20&%20SDG%204-2022/SDG4%E2%80%99s%2010%20targets%20%20Global%20Campaign%20For%20Education.html>
- <https://www.ukfiet.org/2020/post-covid-19-and-sdg4-window-of-opportunity-or-of-opportunism/>
- <https://en.unesco.org/news/united-nations-alert-education-should-be-clear-priority>



THANK YOU